

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First / Second Semester of B. Tech. Examination (All Branches)

May 2011

CL102 Environmental Sciences

Date: 19.05.11, Thursday

Time: 1:30p.m to 04:30p.m

Maximum Marks:70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicate full marks.
3. Make suitable assumptions.

SECTION- I

- Q - 1 (A) Define the environment sciences. Describe the guiding principles of environmental education. (02)
- (B) Describe the structure of atmosphere. (02)
- (C) Describe Eutrophication. (03)

- Q - 2 (A) Explain man – environmental relationship in nature. (04)
- (B) What are the components of physical environment? Explain biosphere. (05)
- (C) Enlist the organic water pollutant. Explain how agricultural practice imparts water pollutants. (05)

OR

- Q - 2 (A) What is environmental lapse rate? With the help of neat sketch explain lithosphere. (04)
- (B) Define the following (Attempt any Five) (05)
- (i) Qualitative pollutant (ii) B.O.D
- (iii) Secondary pollutant (iv) Point source air pollutants
- (v) Biodegradable pollutants (vi) PAN
- (C) What is drinking water quality standard? Why water quality standard needed? Describe Hardness parameter. (05)

- Q - 3 (A) What is global warming? Write its common impact to environment (04)
- (B) Describe turbidity and color as water quality parameter. (05)
- (C) Explain general classification of air pollutants and describe classification according to source of origin. (05)

OR

- Q - 3 (A) Write short note (any two) (14)
- (i) Green house effect and acid rain.
- (ii) Ambient air quality standard
- (iii) Environmental degradation

(P.T.O)

SECTION- II

- Q - 4** (A) Define Following. (02)
(a) Ecology.
(b) Deforestation.
(B) Explain food web with example. (02)
(C) Differentiate energy flow and nutrient flow in an Ecosystem. (03)

- Q - 5** (A) Give classification of natural resources. Why do we conserve natural resources? (04)
(B) Describe importance of biochemical cycle in the biosphere. Describe carbon cycle with Sketch in detail. (05)
(C) What are importance of forest resources for Economic and ecological wealth of a country. (05)

OR

- Q - 5** (A) Explain the structural components of Ecosystem. (04)
(B) What do you mean by Ecological Pyramids? Enlist and explain any one in detail. (05)
(C) Explain advantages and limitation of Non-Conventional energy resources. (05)

- Q - 6** (A) Which are the important theories of population growth? Explain malthusism theory. (04)
(B) How is habitation pattern develop? List all environmental factors governing Human settlement. (05)
(C) Explain environmental problem due to overexploitation of energy resources. (05)

OR

- Q - 6** (A) Explain demographic transition theory. (04)
(B) Enlist methods for population forecasting and explain how overpopulation is acting as pollutants. (05)
(C) Describe in detail uses and overuses of water resources in India. (05)